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Correct Answer

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Practice Exam - 1

Question 79 of 117



A 51-year-old recipient of OHT 5 years ago, with a history of depression, ethanol use and non-adherence to immunosuppressive medications for 6 months was admitted to the hospital in severe cardiogenic shock with an LVEF of <20%. Endomyocardial biopsy shows 3R ACR and AMR 2. He was placed on intravenous heparin and cannulated on VA-ECMO. The patient was treated with IV solumedrol, IV rabbit anti-thymocyte globulin and IV rituximab. The LVEF improved to 35-40% and he was transitioned to high dose oral MMF, tacrolimus, and low dose prednisone. VA-ECMO is uneventfully removed on hospital day 5. On day 12, the patient complains of diffuse joint discomfort and pruritus. He is now febrile and has a new urticarial rash.

Mild new neutropenia and thrombocytopenia are noted on laboratory examination. The patient's spleen is mildly enlarged. Platelet factor 4 antibody testing is negative.

Which is the most likely etiology of these findings?

- A. A re-exacerbation of acute rejection
- ✓ B. A complication of intravenous immunosuppressive agents
- C. Heparin induced thrombocytopenia
- D. An opportunistic infection
- E. A complication of oral immunosuppressive agents

Commentary

References

Testing Point: The Board-certified AHFTC specialist should recognize the complications of immunosuppressive agents commonly used in heart transplant care.

Correct Answer: B. a complication of either intravenous immunosuppressive agents, in this case likely thymoglobulin or rituximab.

Rationale: The patient presents with high grade ACR/AMR complicated by cardiogenic shock. He is treated aggressively for both. While there is some practice variation in management of these cases, both thymoglobulin and rituximab are reasonable treatment options for these forms of high grade rejection. The patient improved with therapy but developed a classic presentation of acute serum sickness. Serum sickness presents with fevers, polyarthralgias, dermatoses (often urticarial in nature), cytopenias and multiorgan failure. The onset is 1-2 weeks after first receipt of the offending agent and can lead to significant morbidity if not recognized and treated promptly. Treatment commonly involves high dose corticosteroids and/or plasmapheresis.

Both thymoglobulin and rituximab are known agents that cause acute serum sickness. Thymoglobulin has the stronger association (some estimates of up to 16% of patients) but it is well described with rituximab as well. The presence of jaw pain is more specific to thymoglobulin. The timing of his presentation does not exclude either drug as the culprit.

Answers A and C: Acute rejection or acute HIT do not explain the findings.

Answer D: While the patient is at risk for opportunistic reactions, no common opportunistic infection would result in the constellation of symptoms described, particularly when his urticarial rash is considered.

Answer E: Acute serum sickness is not a known complication of the patient's chronic oral immunosuppressive regimen.

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